Assignment 5

B23CM1038

April 3, 2026

Repository & Tracking Links

GitHub Branch: https://github.com/Sirin-890/MLOps-B23CM1038/tree/Assignment-5
WandB Project question 1: https://wandb.ai/b23cm1038-prom-iit-rajasthan/Assignment-5-vit_lora?nw=nwuserb23cm1038
WandB Project question 2: https://wandb.ai/b23cm1038-prom-iit-rajasthan/Assignment-5-Adversarial?nw=nwuserb23cm1038
Hugging Face Models: https://huggingface.co/bappaiitj/vit_lora

1 Question 1: ViT-Small Finetuning with LoRA

We utilised a small ViT pretrained on ImageNet and evaluated it on CIFAR-100. We finetuned the classification head solely, and sequentially applied PEFT LoRA into the Attention QKV matrices.

1.1 LoRA Hyperparameter Search

Below are the results for evaluating variations in ranks and alphas (with Dropout=0.1):

LoRA layers	Rank	Alpha	Dropout	Test Accuracy (%)	Trainable Params
Without	N/A	N/A	N/A	79.56	38,500
With	2	2	0.1	89.98	75,364
With	2	4	0.1	89.77	75,364
With	2	8	0.1	89.64	75,364
With	4	2	0.1	90.01	112,228
With	4	4	0.1	90.08	112,228
With	4	8	0.1	90.00	112,228
With	8	2	0.1	89.81	185,956
With	8	4	0.1	**90.28**	185,956
With	8	8	0.1	90.22	185,956

Table 1: Experiment Table spanning combinations of Alpha and Rank. Best configuration (Rank=8, Alpha=4) is bolded.

1.2 Best LoRA Configuration – Training and Validation Metrics

The best hyperparameter configuration (Rank=8, Alpha=4, Dropout=0.1) was identified via Optuna search. Below are the per-epoch training and validation metrics:

Epoch	Train Loss	Val Loss	Train Acc (%)	Val Acc (%)
1	0.6719	0.3826	82.45	87.99
2	0.2754	0.3535	91.32	89.42
3	0.1957	0.3550	93.62	89.51
4	0.1397	0.3554	95.57	89.71
5	0.0975	0.3643	97.04	89.73
6	0.0688	0.3691	98.11	89.95
7	0.0501	0.3786	98.85	89.74
8	0.0382	0.3774	99.27	89.90
9	0.0318	0.3786	99.46	89.83
10	0.0286	0.3790	99.60	89.87

Table 2: Epoch-wise training and validation metrics for the best LoRA configuration (Rank=8, Alpha=4, Dropout=0.1). Experiment: Best_LoRA, Rank: 8, Alpha: 4.

1.3 Observations

All LoRA configurations substantially outperform the baseline (no LoRA: 79.56%), with test accuracies ranging from 89.64% to 90.28%. The best configuration (Rank=8, Alpha=4) achieved 90.28% test accuracy. Training loss steadily decreases from 0.6719 to 0.0286 over 10 epochs, while validation accuracy plateaus around 89–90%, suggesting the model generalises well without overfitting on CIFAR-100.

2 Question 2: Adversarial Attacks (IBM ART)

A ResNet18 model was trained from scratch on CIFAR-10 to achieve robust test accuracy (evaluating at > 72%). FGSM Attacks were iteratively applied. Subsequently, a ResNet34 was tasked to act as a binary adversarial detector for PGD and BIM.

2.1 Analysis of Attacks and Defenses

Clean vs. Adversarial Performance (Scratch vs. ART): The clean ResNet18 achieved a baseline test accuracy of 78.92%. When applying FGSM attacks, we observed a stark contrast between implementing the attack from scratch and utilizing IBM ART. As shown by the specific evaluations, the manual FGSM attack causes an aggressive drop in performance at higher perturbation levels. In contrast, running FGSM via IBM ART resulted in a somewhat more controlled accuracy decay. For instance, at $\epsilon = 0.05$, scratch accuracy collapsed to 18.73% while ART maintained 28.93%.

Perturbation Strength vs. Performance Drop: Increasing the perturbation strength (ϵ) significantly impacted the classification accuracy. With a small $\epsilon = 0.01$, performance dropped moderately (Scratch: 62.67%, ART: 70.90%). By the time ϵ increased to 0.05 and 0.1, the naive scratch implementation caused the model's accuracy to plummet to 18.73% and then 7.39%. However, using IBM ART for the identical ϵ values yielded slightly more robust accuracies of 28.93% and 16.75% respectively. This demonstrates the fragility of undefended networks under standard ϵ thresholds.

FGSM Attack Summary Table

Perturbation (ϵ)	Clean Accuracy (%)	Scratch FGSM Acc (%)	ART FGSM Acc (%)
0 (Baseline)	78.92	—	—
0.01	—	62.67	70.90
0.05	—	18.73	28.93
0.10	—	7.39	16.75

Table 3: Comparison of ResNet18 accuracy under FGSM attack: Scratch implementation vs. IBM ART across varying perturbation strengths ϵ.

2.2 Results

Attack Type	Detector	Detection Accuracy (%)
PGD	ResNet34	$99.91 \geq 70\%$
BIM	ResNet34	$100.00 \geq 70\%$

Table 4: Adversarial Detector Evaluation Metric.

Detection Performance Comparison (PGD vs. BIM): The binary adversarial detectors—employing pretrained ResNet34 architectures to distinguish between clean images and adversarial forms—exhibited exceptional discrimination capabilities, vastly exceeding the $\geq 70\%$ requirement. The BIM detector recorded a perfect accuracy of 100.00%, performing slightly better than the PGD detector which achieved 99.91%.

2.3 Qualitative Examples

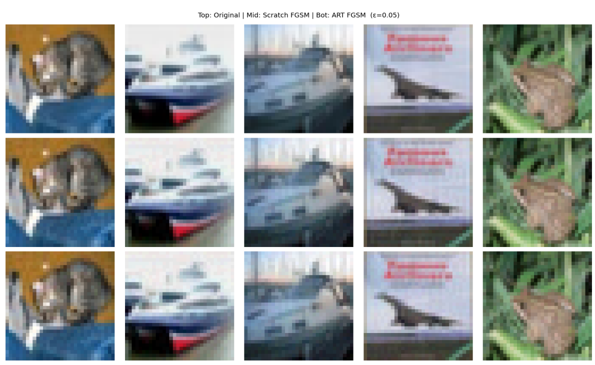

Figure 1: Visual comparison of Original (Top), Scratch FGSM (Mid), and ART FGSM (Bot) adversarial images.

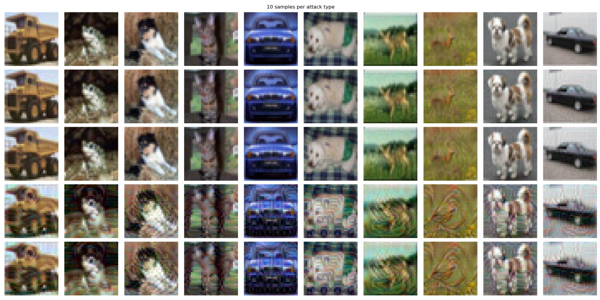

Figure 2: All 10 Samples displaying Clean, FGSM Scratch, FGSM ART, PGD, and BIM generated images.